CUSTOMER SEGMENTATION ANALYSIS USING PYTHON

Data Analytics – Mini Project

JANUARY 1, 2026

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Introduction

Customer segmentation is an important analytical technique used to identify groups of customers with similar characteristics and purchasing behaviour. Organisations frequently use segmentation approaches to improve marketing effectiveness, customer relationship management, and business decision-making. By understanding differences among customers, businesses can allocate resources more effectively and develop targeted strategies that better meet customer needs.

This project utilised the Mall Customers dataset to investigate customer demographics, annual income levels, and spending patterns. Python was used to conduct data preparation, exploratory data analysis (EDA), correlation analysis, and clustering techniques. The aim of the investigation was to identify meaningful customer segments that may support marketing and business planning decisions. Furthermore, the project demonstrated how data analytics techniques can be applied to convert raw customer information into useful business insights.

Data Preparation and Exploration

The dataset consisted of 200 customer records and five variables: Customer ID, Gender, Age, Annual Income, and Spending Score. Before conducting the analysis, the dataset was examined to ensure its suitability for further investigation. Functions such as info(), describe(), isnull().sum(), and duplicated().sum() were used to assess data quality and identify potential inconsistencies.

The results confirmed that the dataset contained no missing values and no duplicate records. Consequently, no extensive cleaning procedures were required. The absence of missing data improved the reliability of the analysis because all customer records could be included without the need for imputation or deletion. According to Chicco, Oneto and Tavazzi (2022), proper data preparation contributes significantly to the quality and reliability of analytical outcomes.

Descriptive statistics were generated to provide an overview of customer characteristics. The analysis showed that customer ages ranged from 18 to 70 years, demonstrating a broad representation of age groups. Annual income values varied considerably across customers, while spending scores exhibited substantial variation, indicating different purchasing behaviours within the customer base.

Several visualisation techniques were applied to support exploratory analysis. Histograms were used to examine the distribution of age, annual income, and spending score variables. A gender distribution chart provided insight into the composition of the customer population, while scatter plots were used to explore potential relationships between annual income and spending behaviour. These visualisations simplified the interpretation of the dataset and highlighted patterns that may not be immediately visible through summary statistics alone. McKinney (2022) explained that visualisation techniques assist analysts in identifying trends, patterns, and irregularities within complex datasets.

Customer Segmentation Analysis

The primary objective of the project was to identify customer groups based on spending behaviour and annual income. To achieve this objective, K-Means clustering was applied. K-Means is an unsupervised machine learning technique that groups observations according to similarities in selected variables.

Prior to clustering, data standardisation was performed using the StandardScaler function. Standardisation ensured that annual income and spending score contributed equally to the clustering process. Without standardisation, variables with larger numerical ranges may exert a greater influence on the clustering outcome.

The Elbow Method was subsequently applied to determine an appropriate number of clusters. This method calculates the Within-Cluster Sum of Squares (WCSS) for different cluster numbers and identifies a point where further increases in cluster quantity produce only limited improvements. Based on the Elbow Method results, five clusters were selected for the final model.

After applying K-Means clustering, customers were assigned to one of five segments according to their income and spending characteristics. A cluster visualisation was generated using scatter plots, enabling clear identification of customer groups and supporting interpretation of the clustering results.

Findings and Discussion

The analysis identified five customer segments with distinct income and spending profiles. Certain customer groups displayed relatively high annual incomes combined with high spending scores. These customers may represent valuable customer segments because they contribute significantly to business revenue and demonstrate strong purchasing activity.

Other clusters contained customers with comparatively high incomes but lower spending scores. This finding suggests that income alone does not necessarily determine purchasing behaviour. Although these customers possess spending capacity, they may not currently be highly engaged with products or services. Consequently, such groups may represent opportunities for future marketing interventions.

Additional customer segments exhibited moderate income and moderate spending characteristics, indicating relatively balanced purchasing behaviour. These groups may represent stable customer categories that contribute consistently to sales performance. Conversely, some customers displayed lower spending scores despite differing income levels, suggesting that behavioural factors beyond income may influence purchasing decisions.

Correlation analysis formed another important part of the investigation. The correlation matrix indicated only limited linear relationships among age, annual income, and spending score variables. This result suggests that customer behaviour is influenced by multiple factors and cannot be fully explained through simple correlations alone. The clustering approach therefore provided deeper insight by identifying behavioural patterns that were not immediately evident through traditional statistical measures.

The visualisation outputs further supported interpretation of the findings. Histograms highlighted variation across demographic and spending variables, while scatter plots revealed clear separation among customer groups. These graphical outputs enhanced understanding of customer behaviour and demonstrated the usefulness of visual analytics in customer-focused investigations.

Recommendations

The findings provide several practical implications for business decision-making. Customers identified within high-income and high-spending segments may benefit from loyalty programmes, exclusive promotions, or personalised marketing campaigns aimed at maintaining engagement and customer satisfaction. Retaining these customers may contribute positively to long-term business performance.

Customers exhibiting high income but comparatively low spending behaviour may represent a valuable growth opportunity. Organisations may consider targeted marketing strategies designed to encourage increased purchasing activity within these groups. Product recommendations, tailored offers, and customer engagement initiatives may support such objectives.

Furthermore, future investigations may benefit from the inclusion of additional variables such as transaction frequency, product preferences, purchase history, and customer satisfaction measures. The incorporation of these factors may improve segmentation accuracy and provide a more comprehensive understanding of customer behaviour.

Regular segmentation exercises should also be conducted because customer behaviour may change over time. Periodic analysis would allow organisations to monitor evolving purchasing patterns and adjust business strategies accordingly.

Conclusion

The project demonstrated the practical application of Python-based analytical techniques in customer segmentation. Exploratory data analysis confirmed that the dataset was complete and suitable for investigation. Visualisation techniques provided insight into customer demographics and spending behaviour, while correlation analysis examined relationships among key variables.

The application of K-Means clustering successfully identified five distinct customer segments based on annual income and spending score. The findings highlighted differences in customer purchasing behaviour and demonstrated that spending activity cannot be explained solely through income characteristics. Overall, the investigation illustrated how customer segmentation can support evidence-based marketing decisions, improve customer understanding, and contribute to strategic business planning.

References

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